

1. 计算下列反常积分:

(1)

$$\int_0^{+\infty} \frac{dx}{e^x + e^{-x}};$$

(2)

$$\int_1^{+\infty} \frac{\arctan x}{x^2} dx;$$

(3)

$$\int_0^{+\infty} e^{-pt} \sin \omega t dt \quad (p > 0, \omega > 0);$$

(4)

$$\int_{-\infty}^{+\infty} \frac{dx}{x^2 + 2x + 2};$$

2. 判断下列无穷积分的敛散性:

1)

$$\int_1^{+\infty} x^\alpha e^{-x} dx;$$

2)

$$\int_0^{+\infty} \frac{x^2}{\sqrt{x^5 + 1}} dx;$$

3. 判别下列瑕积分的收敛性:

(1)

$$\int_1^2 \frac{\sqrt{x}}{\ln x} dx;$$

(2)

$$\int_0^1 \frac{1}{\sqrt{x} \ln x} dx;$$

4. 求下列极限:

(1)

$$\lim_{\alpha \rightarrow 0} \int_{-1}^1 \sqrt{x^2 + \alpha^2} dx;$$

(2)

$$\lim_{\alpha \rightarrow 0} \int_0^2 x^2 \cos \alpha x dx;$$

5. 求 $F'(x)$, 其中:

(1)

$$F(x) = \int_x^{x^2} e^{-y^2} dy;$$

(2)

$$F(x) = \int_{\sin x}^{\cos x} e^{\sqrt{1-y^2}} dy;$$

(3)

$$F(x) = \int_{a+x}^{b+x} \frac{\sin\left(\frac{x}{y}\right)}{y} dy;$$

(4)

$$F(y) = \int_0^1 \ln(x^2 + y^2) dx.$$

6. 用求导法求下列积分：

$$\int_0^\pi \ln(1 - 2a \cos x + a^2) dx \quad (|a| < 1).$$

7. 应用积分交换次序求下列积分：

$$\int_0^1 \sin\left(\ln \frac{1}{x}\right) \frac{x^b - x^a}{\ln x} dx \quad (a > 0, b > 0).$$

8. 证明：欧拉等式

$$\int_0^1 \frac{dx}{\sqrt{1-x^4}} \int_0^1 \frac{x^2 dx}{\sqrt{1-x^4}} = \frac{\pi}{4}.$$